

Integration — Lesson 24 Test: Tangent, Secant & Friends

10 questions · show your work in the box · no calculator needed · exact values (fractions, $\sqrt{}$, π) unless asked for decimals · don't forget + C

Name: _____ Date: _____ Score: _____ / 10

1. Using the definitions $\tan x = \sin x / \cos x$ and $\sec x = 1/\cos x$, find the exact values of $\tan(\pi/3)$ and $\sec(\pi/3)$. 1 pt

2. Find the exact values of $\tan 0$, $\sec 0$, and $\tan(\pi/4)$. Then name one value of x where $\tan x$ is undefined and explain why. 1 pt

3. Find $\int \sec^2 x \, dx$, and check your answer by differentiating. 1 pt

4. Find $\int \csc x \cot x \, dx$. (Careful with the sign!)

1 pt

5. Find $\int (2\sec^2 x + \cos x) \, dx$.

1 pt

6. Compute $\int_0^{\pi/4} \sec^2 x \, dx$.

1 pt

7. Compute $\int_0^{\pi/3} \sec x \tan x \, dx$.

1 pt

8. Find $\int \sec^2(3x) dx$, and check your answer by differentiating.

1 pt

9. A student claims $\int \csc^2 x dx = \cot x + C$. Check the claim by differentiating, say whether it is right or wrong, and give the correct answer if needed.

1 pt

10. Use u-substitution to find $\int \tan x dx$, showing every step (hint: write $\tan x = \sin x / \cos x$ and let $u = \cos x$).

1 pt
